Keivos Shadowing v1 Spec

Oct 27, 2026 · @Tyson

1. Product scope, philosophy, and release boundaries

Keivos Shadowing is a physician-shadowing scheduler owned by Keivos LLC and run at shadow.keivos.com. Each pre-med society or similar group is one organization on the platform; Organization is the product word for it. The BYU-Idaho Pre-Med Society is the first, piloted through Dr. Curtis's office with handpicked students. The six clickable mock-ups (student app, physician portal, coordinator portal, physician onboarding, admin, emails) are the visual reference; this doc is the behavioral one. Where they disagree, this doc wins.

Release boundaries

	Pilot (BYU-Idaho, now)	Commercial release (later)
Organizations	One, provisioned by Keivos	Self-service sign-up, verified by Keivos
Price	Free, outside the guarantee	$29 a month or $229 a year; 90-day guarantee
Student verification	Activation code, plus manual access requests	Same, plus automatic Canvas verification when the integration is proven
Waitlist	Notify everyone	Notify everyone, or sequential offers
Admin roles	Lead and Organization Administrators with permissions	Same
Recruitment and training toolkits	Not required	Included with every subscription
Billing, invoices, refunds	None	Required

Pilotable means every pilot-column item in section 16 is built and running for one organization.

Pricing (commercial release)

Plan	Price	Billing
Monthly	$29	Every month
Annual	$229	Once a year, 34.2% below twelve monthly payments
Introductory guarantee	90 days	First subscription only, either plan

All core features are in both plans. No per-student or per-physician fees and no limits on how many participate. The guarantee refunds eligible payments made in the first 90 days after the first successful payment, with no recruitment quota or minimum usage. The BYU-Idaho pilot stays free and outside the guarantee. Subscriptions renew automatically unless cancelled, disclosed before purchase, with cancellation available from Settings.

Design decisions that stay fixed

- Physicians control their own availability, approvals and participation, and work mainly from email.
- One physician account serves every organization; a physician connects it to a new organization instead of creating a second profile. Physicians never pay.
- Organizations own their program rules, physician relationships and student networks. Keivos owns platform security, infrastructure and institution settings.
- No-shows require manual reinstatement by leadership. Physicians are never told about administrative penalties.
- Student reflections are private and excluded from every analytics query and organization export.
- Students are never told what leadership tracks.

Out of scope entirely

- Letter-of-recommendation tracking, a physician services marketplace, a Keivos-hosted learning management system, roster imports, LTI integration.

2. Roles, permissions, and authentication

Seven roles. Students, administrators, physicians and scheduling coordinators each sign in with an emailed 6-digit code; physicians and coordinators also act from tokened links in email. Every permission is enforced server-side.

Role	Who	Can
Keivos administrator	Keivos staff	Provision and configure organizations (email domains, time zone), billing infrastructure, the specialty list, support
Lead Administrator	Faculty or student leadership; at least one per organization	Everything an organization can do, plus billing, account ownership, comprehensive exports, and every other administrator's role and permissions
Organization Administrator	Leadership with a custom permission set granted by a lead	Only the permission groups granted; sees only those tabs and settings
Student	Verified school account	Find a physician, My shadowing, Account
Physician	The person on the global physician account, verified by their professional email	Their portal: Overview, Schedule, My Organizations, My Profile
Scheduling Coordinator	Office staff with their own free Keivos account, authorized by one or more physicians for specific practice locations	Their portal: Overview, Schedule, Availability, Visits & Requests, Practice locations, Account. Scheduling work for the locations they are authorized for; the rest of the physician's schedule only as busy time

Faculty adviser is a designation on an administrator, not a permission level. An adviser can be a Lead or an Organization Administrator.

Permission groups (administrators)

Permission	Grants
Physicians	Review, approve, pause and resume physicians; read-only profile view; Recruitment view with the invitation link and outreach resources
Students	Eligibility and access requests, cap requests, reinstatement, graduation, revocation
Bookings	View and cancel bookings
Analytics	Analytics, term summary
Program settings	Booking rules, student access, organization profile, academic terms
Administrator invitations	Invite Organization Administrators, within the inviter's own permissions
Data exports	Export the organization's records

Physician versus Scheduling Coordinator

Capability	Physician	Scheduling Coordinator
View the physician's full schedule	Yes	Yes; visits at locations they are not authorized for appear only as Busy, with no student details
View visits at authorized locations	Yes	Yes
Add or edit availability at authorized locations	Yes	Yes; dates they add are offered under the physician's name
Approve or decline student requests	Yes	Only when the physician delegated it for that location
Confirm attendance	Yes	Only when the physician's communication preference routes attendance questions to the location's coordinator
Cancel visits	Yes	Yes, at authorized locations, without delegation

Capability	Physician	Scheduling Coordinator
Edit location logistics, requirements and paperwork	Yes	Yes
Manage the global professional profile	Yes	No
Add or remove organizations, or change their settings	Yes	No
Add, remove or change coordinators	Yes	No
Change the physician account email	Yes	No

Safeguards

- No privilege escalation: an administrator can grant only permissions they hold, and only to new invitees, never by editing existing administrators.
- The last active Lead Administrator cannot be removed or demoted; the control is disabled.
- Granting the lead role, changing the billing contact, and comprehensive exports require a fresh code to the acting administrator's email (step-up).
- Every permission change writes to the audit log and emails the affected administrator and the leads.
- A coordinator's access is granted per physician and per location by that physician alone, and is removed the same way. Removal takes effect on the next request.

Authentication

- Students: school-domain email plus a 6-digit code at sign-up and every sign-in. No passwords.
- Administrators: 6-digit code to an address on the organization's administrator list. Invitations are accepted by signing in with the invited address.
- Physicians: one account keyed by an immutable physician id. The professional email is unique and verified by code, and can be changed without breaking any connection. Portal sign-in is a 6-digit code to that email. A link in any Keivos email opens the portal on a device that already holds a session and asks for the code otherwise.
- Scheduling coordinators: one account keyed by an immutable id, with a verified work email. Sign-in by code; the same tokened links in email as physicians, scoped to the locations they are authorized for.

- Email action links (approve, decline, confirm attendance, cancel, renew, confirm profile, authorize a coordinator): single-use tokens bound to one action, one target and one recipient, carrying the recipient's role. A forwarded link still acts as the recipient it was issued to and never opens a portal session, so a forwarded email gives no lasting control of an account.
- Whether an email already has a Keivos account is never revealed on a public page. A coordinator who enters a physician's email is told only that the physician has been emailed; Keivos decides privately whether that is an authorization request or a confirmation of a prepared profile.
- Codes: 6 digits, 15-minute expiry, 5 attempts. Portal sessions expire after 90 days of silence; each sign-in renews them.

What each role sees

- Students see the physician's card and bio, and for each practice location with open dates its address, meeting point, parking and requirements. Physician and coordinator emails never appear; the day-of contact appears only in the confirmation email. Students see the organization's own name as the label, never "society" or "organization".
- Physicians and coordinators who approve requests see the student's name, year, organization, optional photo and optional bio, and contact details for visits they host. Nothing leadership tracks (strikes, blocks, eligibility history, disciplinary notes) is ever shown to them.
- Administrators open physician profiles read-only, with Pause and Resume as the only actions, and never see another organization's name, rules or bookings.
- Administrators cannot set up a physician. Only the physician or a scheduling coordinator can, from the invitation link.
- Students are never shown what leadership tracks.

3. Data model and multi-organization isolation

Twenty-seven tables. A physician is one Keivos-wide account with an immutable id, a `practice_locations` row per place they host students, a `physician_orgs` row per organization and a `coordinator_authorizations` row per coordinator. Everything else carries `org_id`; row-level security scopes each organization and the database, not the app, prevents double booking. The table stays named `orgs`; "Organization" is the product word for it everywhere.

Names are stored as parts everywhere (first, last, optional suffix, optional preferred display name) and rendered consistently: the display name replaces first and last when set but never overwrites them. Physician credentials are a structured, ordered list (primary MD or DO, then any others) rendered after the name: "Elena Marsh, MD, MPH". Phone numbers are stored as E.164 and formatted for display.

Table	Key fields	Notes
orgs	id, name, affiliation, slug, logo_url, accent_color, welcome_message, contact_email, program_description, student_prep[], participation_terms[], shadowing_enabled, email_domains[] (Keivos), timezone (Keivos, IANA), settings jsonb	settings: cancel_hours, cap_per_month, waitlist_policy (everyone, sequential), grad_access (end, extend), grad_months, elig_renewal (term, year, revoked), allow_requests, activation_code, code_expires. Invitation link is shadow.keivos.com/join/{slug}; program_description, student_prep and participation_terms are shown on it
org_admins	id, org_id, email, first_name, last_name, suffix, display_name, role (lead, admin), adviser, perms[], status (invited, active), invited_by, invited_at, accepted_at, digest_day, alerts[]	The administrator list. alerts holds the optional operational alerts the person chose
terms	id, org_id, name, starts_on, ends_on	A term's window runs from its start to the next term's start. Past terms are read-only

Table	Key fields	Notes
users	id, org_id, email, email_verified_at, first_name, last_name, suffix, display_name, phone (E.164), photo_url, bio (300 chars), year, grad_month, strikes, blocked_at, reinstated_at, graduated_on, access_until, profile_confirmed_term	Students. Identity only; eligibility lives in eligibility_records. Photo and bio are optional and never required to book
eligibility_records	id, user_id, org_id, status (pending, approved, denied, expired, revoked), method (code, manual, canvas), requirement_version, note, requested_at, decided_at, decided_by, expires_at	A pending row is an access request. One current row per student per org; history kept

Table	Key fields	Notes
physicians	id (immutable), email (unique, professional, changeable), email_verified_at, first_name, last_name, suffix, display_name, credentials[] (ordered; first is MD or DO), specialty_id, bio, photo_url, attendance_to (physician, coordinator), reminder_days (1, 2, 7, null), summary_cadence (weekly, every_2_weeks, monthly, off), renewal_reminders, status, paused_until, submitted_by (physician, coordinator), submitted_by_id, confirmed_at, created_at	The global account: identity, professional profile and email preferences only. paused_until null means paused until resumed. A coordinator-prepared account has status pending_confirmation until confirmed_at is set
coordinators	id (immutable), email (unique, work), email_verified_at, first_name, last_name, suffix, display_name, phone, created_at	Scheduling Coordinator accounts. Free; one per person, however many physicians they coordinate for
coordinator_authorizations	id, coordinator_id, physician_id, location_ids[], can_approve, status (invited, requested, active, removed), invited_by (physician, coordinator), created_at, accepted_at	One row per coordinator per physician. invited: the physician added them and they have not accepted; requested: the coordinator asked and the physician has not answered. The physician alone edits location_ids and can_approve

Table	Key fields	Notes
practice_locations	id, physician_id, name, address, city, state, zip, timezone (IANA), meet_at, parking, dayof_contact, requirements[], paperwork jsonb (mode, statement, links[], files[]), is_default, created_at	Reusable across organizations. Exactly one default per physician; rules, dates, slots and bookings point at a location. Who coordinates a location comes from coordinator_authorizations
physician_orgs	id, physician_id, org_id, status (pending, active, paused), paused_until, years_accepted (any, juniors_seniors), approve_each, prefs_set, availability_mode (shared, own), connected_at, connected_via (invitation, coordinator, portal_code), renewed_for_term	One row per organization. Preferences are per organization; prefs_set false means the physician skipped them and students cannot book yet
org_recommendations	id, physician_id, org_name, contact_name, contact_email, note, sent_at, status (sent, joined)	Recommend Keivos to an Organization. Keivos sends the email; the physician is told when that organization joins
specialties	id, name, added_by_physician_id, created_at	One Keivos-wide list, seeded with 45. Other on the profile step adds the typed name at once

Table	Key fields	Notes
availability_rules	id, physician_id, location_id, org_id nullable, created_by (physician or coordinator id), repeat (daily, weekly, every_2_weeks, monthly), weekday, nth, start_time, end_time, starts_on, ends_on, lead_days, horizon_days, students_per_slot (1 to 50)	org_id null means shared availability (the default); set, the rule is offered to that organization alone
manual_dates	id, physician_id, location_id, org_id nullable, created_by, date, start_time, end_time, lead_days, horizon_days, students_per_slot (1 to 50)	Individual dates
slots	id, physician_id, location_id, org_id nullable, source, rule_id or manual_id, date, start_at, end_at, capacity, status (open, closed, removed)	Unique on (physician_id, start_at); an exclusion constraint also forbids overlapping intervals per physician. Times are stored in UTC from the location's time zone

Table	Key fields	Notes
bookings	id, org_id, slot_id, location_id, seat, student_id, physician_id, status (queued, requested, confirmed, declined, auto_declined, withdrawn, cancelled_student, cancelled_physician, completed, no_show), queue_position, requested_at, decided_at, decided_by, late, late_note, created_at, cancelled_at, cancelled_by	One row per student per slot, ever: a partial unique index on (slot_id, student_id) covers every status except withdrawn and cancelled, so a declined student cannot request the same slot again. seat 1 to capacity on confirmed rows, with a partial unique index on (slot_id, seat). An exclusion constraint per student forbids overlapping (date, time) ranges across rows in queued, requested or confirmed. Booking and approval run in one transaction that locks the slot row. location_id is copied from the slot so history survives later location edits
visit_logs	id, booking_id, hours, submitted_at, reflection fields, reflections_updated_at, attendance (pending, confirmed, denied), attendance_at, attendance_by	Hours written once; reflections editable; private to the student
waitlist	id, student_id, physician_id, org_id, created_at, status (waiting, notified, offered, booked, expired), notified_at, offer_expires_at	offered and offer_expires_at serve the sequential policy
physician_blocks	id, physician_id, student_id, created_at, created_via, note	Physician-wide; the student is never told

Table	Key fields	Notes
cap_requests	id, student_id, org_id, month, note, requested_at, status, extra_visits, decided_by, decided_at	
activity_log	id, physician_id, actor_type (physician, coordinator, admin, system), actor_id, action, target_id, created_at	Physician-level history; coordinator actions are attributed to the coordinator
audit_log	id, org_id, actor_type, actor_id, action, target, details jsonb, created_at	Permission changes, eligibility decisions, settings changes, billing, exports, sign-ins
email_tokens	id, token_hash, action, target_id, recipient_email, role (student, admin, physician, coordinator), location_id nullable, expires_at, used_at	A coordinator token carries the location it may act on
email_codes	id, email, code_hash, purpose, org_id nullable, expires_at, used_at, attempts	
email_log	id, org_id nullable, type, to, booking_id, physician_id, sent_at, status	Bounces alert the responsible admins; physician and coordinator account mail has no org
subscriptions	id, org_id, plan (monthly, annual, pilot), state (trialing, active, past_due, cancelled), started_at, renews_at, guarantee_until, billing_email, provider_customer_id, provider_subscription_id	Commercial release

Table	Key fields	Notes
invoices	id, org_id, subscription_id, amount, status, issued_at, paid_at, pdf_url	Commercial release

Keivos-controlled versus organization-editable

Keivos sets, organization sees read-only	Organization edits
Email domains, time zone, slug and invitation link, plan, enabled flag, specialty list	Name, affiliation, logo, accent colour, welcome message, program description, what every student has, what participation involves, contact email, booking rules, student access policies, terms, administrators, billing contact

Where each physician field lives

Field	Lives on
Name parts, credentials, specialty, professional email, photo, bio, attendance routing, reminder timing, summary cadence, renewal reminders, Schedule Pause	physicians (global)
Name, address, time zone, meet at, parking, day-of contact, requirements, site paperwork, default flag	practice_locations (global, reusable)
Who coordinates a location, and whether they may approve requests	coordinator_authorizations (global; the physician's to change)
Rules and individual dates, each at one location	availability_rules and manual_dates; shared when org_id is null, own when set
Status, pause for this organization, renewal, connected date, Which students, Approve each, preferences set, availability mode	physician_orgs

Field	Lives on
Bookings, waitlist, cap requests, eligibility	per org
Blocked students	physician_blocks, physician-wide

One time, one visit

- A slot is one physical visit at one location and belongs to the physician. A shared rule's slot is visible to every active organization connection; a seat goes to the first confirmed booking from any organization. An own rule's slot is visible to one organization.
- Slots never overlap: unique on (physician_id, start_at) and an exclusion constraint on the time range. Saving a rule or date whose times overlap an existing slot from any other rule is refused with "Overlaps your availability on <day>".
- A student never holds two overlapping bookings. The exclusion constraint covers queued and requested rows as well as confirmed ones, so a student cannot request two visits that overlap even with different physicians; the app checks when it shows dates, when the request is submitted, and again when a physician approves it.
- Seats are filled only by approval or automatic confirmation inside the slot transaction; a request queue never holds a seat. Two simultaneous confirmations for the last seat cannot both succeed.
- A booking's address, meeting point, parking, day-of contact, requirements, paperwork and coordinator come from the slot's location at the time of booking.
- An organization's export or screen never includes another organization's students, bookings, settings or name.
- Store all timestamps in UTC; render students and administrators in org.timezone, and the physician and coordinator portals in the location's time zone. No field holds patient information.

4. Student onboarding and eligibility verification

Identity, eligibility and booking authorization are three separate checks. A student can be verified and still unable to book because of a no-show or an expired term.

Flow

1. Register: first name, last name, suffix and preferred name (both optional), school email (must match an org email domain), phone, year in school (Freshman to Senior), expected graduation month, and optionally a photo and a bio of up to 300 characters

("Tell participating physicians a little about your interests and what you hope to learn from shadowing."). Photo and bio are editable from Account and never required to book. Verify the email with a 6-digit code.

2. Eligibility check. Pilot: the student enters the term's activation code, given to them by leadership once they finished the prerequisites. Any term: a student without a code can submit an access request with a note, when the organization allows requests.
3. Verified: activated at once, welcome screen with the organization's welcome message. Not verified: a pending screen ("Awaiting approval"), an email when leadership decides, and the same status on every sign-in until then.
4. Commercial release, when proven: automatic verification of the student's own Canvas completion through an authorized integration (OAuth or LTI 1.3; not described as working until tested). The manual request stays as the fallback. No roster imports, no Keivos-hosted assessments.

The interface is one call, `isEligible(user, org)`, so a verification method can be swapped without touching anything else.

Eligibility states

State	Meaning	Student sees
pending	Access request awaiting a decision	Awaiting approval screen; email on decision
approved	Verified by code, manual approval or Canvas; `expires_at` set by the renewal policy	Normal app; Account shows "Approved <date> · through <date>"
denied	Leadership declined	Email; can request again
expired	`expires_at` passed	Activation step again, with the new code or a request
revoked	Leadership revoked it	Activation step again

Renewal policy (Settings → Student access; default each academic term)

Option	Behavior
Each academic term	Approval expires when the term's window ends; students enter the new term's code or request access again. Two weeks before, they are emailed
Annually	Approval lasts 12 months
Until revoked	Approval lasts until leadership revokes it or another restriction applies

- Expiration prevents new bookings only. Existing confirmed visits stay and are flagged on the Students tab for leadership review.
- Eligibility is independent of no-show suspension; both must be clear to book.
- Term rollover never preserves eligibility that should have expired; the nightly job expires records by `expires_at`, not by term change.
- Admin generates a new activation code each term; the old one stops at once. Revocation and denial are recorded with who decided and when.
- Graduation: `access_until` is the end of the graduation month, or that plus the organization's extension when "Allow continued access" is on. Graduates who lose their school email recover their account through leadership (open decision 17).

5. Physician onboarding, portal, coordinators, and organizations

One account. A physician has one Keivos account, keyed by an immutable id and signed in with their professional email, that connects to any number of organizations. Profile, practice locations and availability are entered once and reused; each organization gets its own settings. Physicians never pay and never need an organization to hold an account.

Entry points

Entry	Who	What happens
Organization invitation link, shadow.keivos.com/join/{slug}	A physician, from the recruitment email or the administrator's Copy invitation link	A landing page with the organization's name, affiliation and program description asks "Do you already have a Keivos physician profile?" with two cards: Use my existing profile, Create a new profile. Below: a line for scheduling coordinators and office staff, and a line for physicians without an invitation
Scheduling coordinator, same link	Office staff setting a physician up	Creates or signs in to their own coordinator account, prepares the physician's setup; the physician confirms by email before anything goes live
Independent registration, shadow.keivos.com/join	A physician with no invitation	Creates the account. No students see them until an organization connects; the done screen offers Recommend Keivos to an Organization
Recommend Keivos to an Organization	Any physician, from My Organizations or the independent done screen	Organization name, contact name, contact email, optional note. Keivos emails the contact (section 10); the physician is told when that organization joins

Create a new profile (seven steps). A step bar shows progress; Back keeps everything typed. Only the first two steps are required to hold an account; every later step has a Skip for now link with one line saying what skipping means. Nothing warns again; the portal's setup checklist carries what is left.

Step	Fields	Skippable	Notes
1 Verify identity	Professional email, 6-digit code	No	An email that already has an account is recognised: "You already have a profile. Sign in to use it."
2 Professional profile	Photo (optional), first name, last name, suffix (optional), preferred display name (optional), primary credential (MD or DO), + Add credential (repeatable, unlimited), specialty (shared list plus Other, which adds the typed name), short bio (optional)	No	Name, specialty and the primary credential are required. The display name is generated: "Elena Marsh, MD, MPH"
3 Connect	The organization's card: program description, what every student has, what participation involves; Availability they see (existing profiles only); "I'd like to connect my Keivos profile to <organization>"	No when invited (independent registration has no Connect step)	Creates a pending physician_orgs row that the organization approves

Step	Fields	Skippable	Notes
4 Practice location	Practice or hospital; street address, city, state, ZIP; where the student should go on arrival; parking; day-of contact; before the student comes (one per line); site paperwork (None, or Required with statement, link, file); invite a scheduling coordinator (optional: name, email, may approve requests)	Yes	Becomes the default location. Time zone defaults from the organization and is editable on the location. More locations are added in the portal
5 Shadowing preferences	Which students (Any year, Juniors & seniors); Approve each student by name (default No)	Yes	Per organization. Skipping leaves prefs_set false: defaults show, but students cannot book until the physician confirms them in My Organizations

Step	Fields	Skippable	Notes
6 Communication preferences	Regular shadowing summary (default on, weekly; cadence Every week, Every 2 weeks or Every month; sent only when there is something to report); term renewal reminders (default on; turning it off shows that unrenewed availability pauses at term start); visit reminder (1, 2 or 7 days before, or none); attendance questions go to (me, the location's scheduling coordinator)	Yes	Skipping applies the defaults shown on the step
7 Availability	Recurring availability, A one-time date, or I'll add availability later, with the rule and date fields from section 6	Yes	Students cannot see the physician until at least one date exists

Done screen: "You're connected to <organization>." with the emails to expect and Open my portal. When anything required was skipped, the done screen shows the setup checklist instead of a warning. Only day-of contact details are ever shared with students.

Setup checklist. A card at the top of the portal's Overview, "Finish setting up shadowing", with a progress bar and four items: Verify your professional email, Create your professional profile, Add a practice location, Set your shadowing preferences. Each unfinished item says why it matters ("Required before offering shadowing visits") and has a Finish button to the right tab. The card ends with: "Your profile is saved. Students won't be able to book shadowing visits until your required setup is complete and you've added

availability." The card disappears once the four items are done, even when the calendar is empty: a complete profile with no availability is a physician taking a break, not an incomplete one.

Use my existing profile (three steps). Verify identity by code to the account email; Connect: the organization's card and Availability they see (My existing availability, listed; or Dates I add for this organization only); Shadowing preferences for this organization (skippable). Creates a pending physician_orgs row that the organization approves like a new physician. No second account is created. The same steps run from the portal's Connect with an organization box when a physician pastes an invitation link or code.

Scheduling Coordinator setup (coordinator-initiated). Steps: Your account (first name, last name, work email, 6-digit code; an existing coordinator account signs in the same way); The physician (first name, last name, professional email, specialty); Practice location (skippable); Proposed availability (recurring, one-time, or Leave it to the physician). The page never says whether the physician's email already has an account. On submit Keivos decides privately: an existing physician receives "<Coordinator> asks to coordinate scheduling for you" with Approve (choose the location and whether they may approve requests) and Decline; a new physician receives "Please confirm your shadowing profile" with one button that verifies the professional email, shows what was prepared and asks them to confirm participation and the organization connection, and a second that declines. Until the physician answers, the account is pending_confirmation, nothing is visible to students or administrators, and the coordinator has no access to that physician. After confirmation the coordinator is authorized for that location, the organization reviews the connection, and the physician's first portal sign-in asks for communication preferences. Unanswered after 7 days: one reminder; after 30 days a prepared profile is deleted and a pending authorization request expires.

Scheduling Coordinator setup (physician-initiated). From a practice location card in My Profile, or on the Practice location step of onboarding, the physician enters a coordinator's name and email and whether they may approve student requests. Keivos emails "<Physician> invited you to coordinate scheduling on Keivos": an existing coordinator account gains the authorization on sign-in; a new person creates one (first name, last name, work email, code). A physician can authorize several coordinators, each for any subset of their locations; a coordinator can serve several physicians.

Independent registration. Steps 1, 2, 4 and 6. The done screen says that students see the profile only once an organization connects, and offers Recommend Keivos to an Organization.

Physician portal. shadow.keivos.com/portal, opened by code or from any Keivos email. Header: name, Sign out. Status (Active, Pending, Paused, Paused until a date) sits on Schedule with Pause or Resume. Four tabs; every edit is the physician's. Upcoming always means confirmed visits; a request is never counted as a visit until it is approved.

Tab	Holds
Overview	Welcome line; the setup checklist while anything required is missing; tiles for upcoming (confirmed) visits, actions needed, open dates and status. Action needed as individual cards, one per item: the active request on a slot (student, year, bio, date, location, organization, seat n of N and how many more are in line; Approve, Decline), an attendance question (Attended, Did not attend), a renewal when due (Keep the same schedule, Change it). Upcoming visits as cards (the next three confirmed, with a link to Schedule when there are more)

Tab	Holds
Schedule	Month calendar with open, requested and booked slots, each naming its location and, when offered to one organization, that organization; a slot with requests shows "N requests", a slot with confirmed students shows the student (or "N students"); click a slot for Remove, or Cancel per confirmed student; + Recurring availability and + Individual date, each asking Applies to (Shared availability by default, or one organization only) and a Location when the physician has more than one; lists of rules and dates with Edit and Remove. Upcoming visits as cards: confirmed only, with student name, year, email, phone, location, organization and Cancel. Requests as cards, one per request in line: the active request with Approve and Decline, each later one marked In line with "Sent to you when the request ahead is answered". History as a section collapsed by default to its title and counts; opened, one card per past, cancelled, declined or withdrawn booking with hours logged, status, Attended and Did not attend when awaiting confirmation, and Block
My Organizations	One card per organization: connected date, status, Availability they see (shared or own), Which students, Approve each, Pause for this organization or Resume, blocked students with Unblock; Recommend Keivos to an organization; Connect with an organization (paste an invitation link or code)

Tab	Holds
My Profile	Professional profile, open by default: photo, first name, last name, suffix, preferred display name, credentials (primary MD or DO select, additional credentials as repeatable fields with + Add credential and remove), specialty, professional email (Change sends a code to the new address), bio. Three sections collapsed by default to a title and one-line summary: Practice locations (count and default; opened, cards with every field, Add, Edit, Make default, Remove, which is refused while the location has future dates, and a Scheduling coordinators block per location listing each coordinator with whether they approve requests and whether attendance questions reach them, Let them approve or Stop approving, Remove, and + Invite a coordinator), Email preferences (summary cadence, renewal reminders, reminder timing; opened, the step 6 fields), Recent activity (last action; opened, the last 50 actions and who did each, coordinators by name)

- Pause opens a dialog: Until resumed (default) or Until a date. The Schedule Pause applies to every organization; each organization card has its own.
- Administrators open the profile read-only from the Physicians tab: four tiles (hours hosted, visits, open dates, students blocked), then real tabs, Profile by default, then Locations, Availability, Students, Blocked and Activity, each showing only its own section, without edit controls, Pause and Resume only. Students shows confirmed visits and requests in line separately. Other organizations are never named, and rules or dates offered to another organization alone are not shown.
- The portal stays email-first: every action an email button offers works without signing in, and the portal is where a physician goes to change something rather than a place they must check.
- Every change writes to activity_log.

Scheduling Coordinator portal. shadow.keivos.com/coordinator, opened by code or from any Keivos email sent to the coordinator. Header: name, Sign out. Everything is scoped to the physicians and locations in the coordinator's active authorizations; the rest of a physician's schedule appears only as busy time. Six tabs.

Tab	Holds
Overview	Welcome line; tiles for upcoming visits at their locations, actions needed, physicians they coordinate for, open dates at their locations. Action needed cards across every physician: requests where approval is delegated (Approve, Decline) and attendance questions for physicians whose preference routes them to the coordinator (Attended, Did not attend); lines noting requests that wait on a physician who approves personally and attendance questions that go to the physician. Upcoming visits as cards, each naming the physician and location
Schedule	A physician selector when they coordinate for more than one; a line naming their locations; the physician's full month calendar. Slots at their locations behave as on the physician's Schedule (student names, Cancel per confirmed student, Remove); slots at other locations render as a hatched Busy block with the time and location in the tooltip and nothing else. Rule and date lists show every rule, with other locations marked and without controls
Availability	The same calendar with + Recurring availability and + Individual date, Applies to and the Location choice limited to their locations, and Edit and Remove on rules and dates at those locations. Dates they add are offered under the physician's name and logged to the coordinator
Visits & Requests	Requests (Approve and Decline when delegated, otherwise "<Physician> approves requests personally"), Attendance to confirm (only for physicians who route attendance questions to the coordinator; others answer personally and can change that in their email preferences), Upcoming visits with Cancel (never delegated; any coordinator of the location may cancel), and a collapsed History table for their locations

Tab	Holds
Practice locations	One block per physician with a card per authorized location: every field, Edit (the location fields of section 5, including paperwork) and the coordinators listed read-only. No Make default, Remove or coordinator changes
Account	Their own name parts and phone; the list of physicians they coordinate for with locations, since date and whether they may approve; Accept on pending invitations; a link to start a physician's profile

- Routine work stays in email: confirmations and reminders for a location go to its coordinators; attendance questions go to them only when the physician's preference says so; the request email goes to a coordinator too when approval is delegated. Every email button works without signing in.
- A coordinator cannot reach the physician's profile, organizations, coordinators or email; those pages do not exist in their portal and their tokens cannot open them.

6. Availability and booking rules

Physicians set availability two ways, on Schedule or during onboarding: recurring rules and individual dates. Both produce slots, and every rule, date and slot belongs to one practice location (the default when the physician has only one). A visit is exactly the hours the physician set for that date; the product has no visit length of its own, and organization copy must not state one. Nobody at the organization enters dates.

Recurring rule. Applies to (Shared availability, the default, or one connected organization only; shown once the physician has a connection), Location (shown when the physician has more than one), Repeat (Daily, Weekly, Every 2 weeks, Monthly), Day of week (hidden for Daily), Which (1st to 4th or Last; Monthly only), Start and End time, Starting (optional; anchors Every 2 weeks), Until (optional), Minimum notice (days), Available in advance (days), Students per visit (1 or 2). Several per physician; each has Edit and Remove with confirmation.

Repeat	Generates
Daily	Every weekday, Monday to Friday
Weekly	That weekday every week from Starting (or today)
Every 2 weeks	That weekday every other week, anchored on Starting (or the first occurrence from today)
Monthly	The nth weekday of each month; Last means the final one

Individual date. One date at one location, with the same Applies to choice, Start, End, Minimum notice, Available in advance and Students per visit. Edit and Remove.

Slot generation

1. Nightly, and immediately after any rule, date or pause change: walk each active rule from today through its Available-in-advance window and create any slot that does not already exist.
2. Slots within Minimum notice of today show to students as "Inside notice". They are kept, not deleted.
3. On a rule change, regenerate future rule slots with no active booking. Never touch a slot with a requested or confirmed booking.
4. Removing a slot on the calendar sets it to removed after confirmation. A booked slot runs physician cancel instead.
5. Pause hides open slots until resumed or until a chosen date. Booked slots stay unless cancelled.
6. Term renewal unanswered by term start (or never asked, because the physician turned renewal reminders off) pauses that physician_orgs row. Records are never deleted.
7. When a slot opens (new, cancelled, unpaused), run the waitlist policy.
8. Overlap is refused at save time and by the database (section 3).

Booking rules. Every rule is a setting on the org, the physician or the organization connection. The student's Account panel lists them in plain words.

Rule	Behavior
Who can book	Student with approved, unexpired eligibility; not blocked; access_until not passed; under the monthly cap; allowed by the physician's year rule; holding no confirmed visit or request in line that overlaps the slot's time
Year rule	Any year, or Juniors & seniors. A physician whose rule excludes the student is hidden entirely
Graduation access	End at graduation (default): booking ends at the end of the graduation month. Allow continued access: graduates keep booking for the organization's set number of months, as long as eligibility is current. Either way `access_until` is an explicit date, admin can mark graduation earlier or undo it, and history and export stay
Monthly cap	cap_per_month requested-or-confirmed bookings per calendar month, default 2, plus extras an admin granted for that month. At the cap, "Request more" opens an inline note; the request becomes an action item and the student is emailed the decision
Minimum notice	Slot date minus today must be at least the rule's or date's lead days
Capacity	Students per visit: a number field, default 1, up to 50 (a technical ceiling, not a policy). Enforced in the database by the slot-row lock and the (slot_id, seat) index. Lowering capacity below the number of confirmed students is refused until the physician cancels specific visits; nothing is cancelled silently

Rule	Behavior
Approve-each physicians	Students join one ordered line per visit, in the order the server records their requests. Before sending, the student is told how many are ahead ("3 students have already requested this visit. You would join the line as #4.") and afterwards sees their place and whether their request is with the physician now. The physician, or a coordinator with delegated approval, sees one active request per open seat and answers it by email or in the portal; approving fills the next seat and the next student in line becomes the active request while seats remain. The request email names the seat ("Seat 2 of 3") and how many wait behind. Requests never expire and are never reassigned because the physician is slow; that is the physician's call. When the visit starts (its start time, not midnight), every request still in line is declined automatically, each student is told nothing is held against them, and the physician gets an informational summary: seats, how many were approved, how many were still waiting. Requests already in line stay valid after the minimum notice passes; the notice only stops new ones. Once every seat is filled, later requests stay in line for a cancellation and no further request is sent to the physician until a seat frees. A waiting student sees only the place they joined at (queue_position, fixed when they joined), never how many seats are filled or who is ahead; seat counts are shown only on auto-confirm visits
Student cancel	Allowed at least cancel_hours before start, after a confirmation dialog
Late cancellation	Inside the window the button reads "Cancel late" and opens a dialog that says it counts as a no-show, with an optional note to leadership. On confirm: booking cancelled, strike +1, blocked_at set, the office told only that the visit is cancelled, leadership emailed with the note

Rule	Behavior
Physician cancel	Any time from the calendar or an email, after confirmation. Student emailed, date reopens
Admin cancel	Any booking from the Bookings tab, after confirmation. Both sides emailed
Completion	When the slot's end passes and the booking is confirmed, status becomes completed and the post-visit prompt is queued
Hours	Logged once by the student (default the slot length), never editable; count once attendance is confirmed
Reflections	Three optional prompts plus a free field; editable any time; private
Attendance	Confirmed from email (Yes, No) or a portal (Attended, Did not attend) by the physician, or by a coordinator of that location when the physician's preference routes attendance questions there; the physician can always answer too. Did not attend asks for confirmation
No-show	Set by a No, a late cancellation, or admin. Effects: logged hours discarded, strike +1, booking access suspended at once, admin action item. No automatic lift, no configurable durations
Reinstatement	Manual only, by an administrator with the Students permission, possibly immediately. The strike stays in history. Physicians are never told about penalties
Student blocked view	Find a physician replaced by a notice: what was recorded, the leadership contact email, that reinstatement can be immediate, that the no-show stays in history, and that physicians are not told. History stays visible

Rule	Behavior
Waitlist	When a physician has no bookable dates, the student can join that physician's waitlist. Policy per organization: Notify everyone (default) emails every waitlisted student when a date opens and the first to book wins; Sequential offers (commercial release) holds the date for one student at a time with a configurable response window, a maximum number of offers before general release, and release to everyone when the date nears the physician's minimum notice. Entries expire at term end
Physician block	From the profile (with confirmation) or the No confirm page. The physician disappears from that student's view; the student is never told; confirmed visits stay unless cancelled
Site paperwork	Shown on the card, detail and confirmation; the booking is held until the site clears the student
Declined requests	A declined request is marked Declined for good: the student cannot request that visit again (enforced by the one-row-per-student-per-slot index), the next student in line becomes the active request if a seat is open, and the student may request any other date with that physician. No reconsideration or appeal feature.
Withdrawn and cancelled requests	A student may withdraw a request from My shadowing at any time; the next in line moves up. When a confirmed student cancels, or the physician cancels one student, the freed seat goes to the next request in line, which becomes active, provided the visit has not started. A student who becomes ineligible, graduates or is blocked leaves the line silently. A physician pause hides open seats but leaves existing lines untouched.

Rule	Behavior
Overlapping visits	A student cannot book or request a visit that overlaps a visit they already hold, confirmed or in line, with any physician. The date shows disabled with "Conflicts with your existing visit". The rule is checked when dates are shown, when the request is submitted, and again when a physician approves (the student's circumstances may have changed); the database exclusion constraint is the final guard. A conflict at submit shows a plain explanation with a link to the existing booking.

- A cancelled booking never reaches the physician's attendance email.
- Every destructive or irreversible action shows a one-button confirmation naming what will happen and who is emailed.

7. Student experience and shadowing history

Two tabs after onboarding, Find a physician and My shadowing, plus an Account popover. No help markers; the popover carries the rules.

Screen	What it does
Onboarding	Step bar Profile (name parts, contact, year, graduation, optional photo and bio) → Verify email → Activation code. Code step offers "Don't have a code? Request access" when the organization allows it; the request takes a note and lands on an Awaiting-approval screen. Done screen shows the organization's welcome message, the cap and the cancellation rule
Sign in	School email, then a 6-digit code. A pending student lands on the awaiting-approval screen; an expired one on the activation step

Screen	What it does
Find a physician	A Physicians / Availability switcher sits above the results; Physicians is the default and the last-used view is remembered on the device. Both views share the filter box (search, specialty checkboxes with counts including empty ones, day chips, a from-to time range, availability), the same hidden-physician rules and the same slots. Physicians view: cards with photo or initials, specialty, site and city, next open date, paperwork flag. Availability view: every bookable date across the matching physicians as its own card in date order, grouped by month, showing date, time, physician, specialty, site and any 2-students, approves-each or site-paperwork note, with a chevron and no button; the whole card is clickable and opens that physician's page with the date preselected, so nothing is committed until the student confirms there. The day and time filters apply to each date in this view. Empty results: "No physicians match these filters." or "No open dates match these filters." A profile-confirm banner once per term
Physician detail	Header with specialty, the location (or "2 locations") and city, bio, then for each practice location that has open dates: address, where to meet, parking, day-of contact ("Shared in your confirmation email"), before you come, paperwork block; approves-each note; open dates as cards: open (seats left on auto-confirm visits; how many are in line, and never the seat count, when the physician approves by name), Booked, Inside notice, Conflicts with your existing visit, Declined, and the student's own date (Yours, or Your request with its place in line), each card naming its location when there is more than one; Join waitlist

Screen	What it does
Confirm booking	When, with, address, cancel-by time; for a physician who approves by name, a line notice (how many are ahead and the place the student would take; no seat count) and a note that the place survives the minimum notice; rules list, acknowledgment; Confirm booking, Send request, or Join the line
My shadowing	Tiles: confirmed hours (and awaiting), visits, upcoming (confirmed, with pending requests counted separately), no-shows. Upcoming (confirmed only) with address, paperwork flag, free-cancellation deadline, Cancel or Cancel late. Requests with the place the student joined at (it never moves) and Withdraw. History (completed, cancelled, declined, withdrawn, not answered in time) with Log hours once, then Edit or Add reflection; Export
Account popover	Profile view (photo, name, bio) and edit (name parts, phone, year, graduation, photo, bio; email locked), Access line ("Approved <date> · through <date>" or the state), Sign out, then Rules and terms: monthly limit, cancellation window, no-show, minimum notice, open dates, booked, approves each student, request line, declined requests, overlapping visits, waitlist, site paperwork, hours, contact info, reflections, access renewal
Blocked state	Find a physician replaced by the notice described in section 6; My shadowing stays

- The organization's own name appears in the header and in every email to students; "society" and "organization" never appear as labels in the student app.
- Export: the student's own hours log with reflections as CSV and PDF. Reflections appear nowhere else.

8. Organization administration and settings

Six tabs. An administrator sees only the tabs and settings their permissions cover; a lead sees everything. Every table sorts by any column and filters by the controls above it; period filters list the current term first, then This year, then All time, then past terms newest first.

Tab	Contents	Actions (permission)
Overview	Tiles: active physicians, open dates, visits this month, activated students, waitlisted, physicians awaiting review, attendance unconfirmed. Action items: pending physician connections (new profile, existing profile or confirmed coordinator setup), student access requests, cap requests, stale attendance, blocked students. Shows the admin's own digest day	Approve, Review (Physicians); Approve, Deny, Grant n, Deny, Reinstate (Students); Resend (Physicians)
Physicians	A Physicians / Recruitment switcher at the top. Physicians view: filters status, specialty, search; columns Physician (with blocked count), Specialty, Site (or the number of locations, with cities), Available (open bookable dates), Booked, Joined (physician or coordinator, with the date), Status. Recruitment view: the invitation link with Copy, and the outreach resources of section 11 with Edit and Download	Approve, Pause dialog, Resume, Profile read-only, Copy invitation link, resource Edit and Download (Physicians)
Bookings	Filters: period, status, physician, search	Cancel with confirmation (Bookings); CSV (Data exports)

Tab	Contents	Actions (permission)
Students	Filters: period, status (Can book, Awaiting approval, Blocked, Not activated, Graduated), year, search. Columns: Student, Year, Graduates, Eligibility (method, approved and expiry dates, or requested, expired, revoked, denied), Visits, Hours, No-shows, Status chips	Actions panel per row: Approve or Deny a request, cap Grant or Deny, Reinstate, Mark graduated, Revoke eligibility, Undo graduation (Students)
Analytics	Period selector, Term summary, three sections	(Analytics)
Settings	Six sections below	(Program settings; leads for Leadership and Subscription)

Settings sections

Section	Holds
Booking rules	Visits per student per month; cancellation window (hours); waitlist policy (Notify everyone, Sequential offers greyed until the commercial release); graduation access (End at graduation, Allow continued access plus months)
Student access	Eligibility renewal (Each academic term, Annually, Until revoked); verification method (Activation code; Canvas shown as not yet available); allow access requests; activation code with Copy and Generate new code (confirmation); approved email domains, locked

Section	Holds
Organization profile	Organization name; University; logo (PNG or SVG, at least 256 px square, under 1 MB; initials as fallback); accent colour (must keep 4.5:1 contrast on white, lighter colours darkened automatically; core layout and navigation unaffected); leadership contact email; time zone, locked; welcome message; program description, what every student has, and what participation involves (shown to physicians on the invitation page and the Connect step)
Leadership and permissions	Administrator table: name, email, Faculty adviser badge, role (editable by leads), permission chips (editable by leads), status (Active or Invited with Resend), digest day (own row only), Remove (disabled for the last lead). Invite form: email, role, adviser, permissions within the inviter's own. My notifications: digest day and optional alert checkboxes, with a note that invitations, permission changes, sign-in and billing notices are always sent
Academic terms	Current and upcoming terms with editable dates, Add term, Remove with confirmation, past terms collapsed and read-only
Subscription and data	Plan, since, billing contact (Change requires step-up), invoices, renewal; export package (section 9) with step-up; retention statement

- Every settings section is collapsed by default to its title and a one-line summary of its current values (for example "2 visits a month · 72-hour cancellation · waitlist: notify everyone"); clicking the header opens it, and several can be open at once.
- Locked settings show the value and "Managed by Keivos. Contact Keivos Support to change it."
- A term's window runs from its start until the next term's start; This year is the calendar year.
- Every settings change, permission change and export writes to `audit_log`.

9. Analytics, reporting, and data exports

Everything derives from bookings, visit logs, slots and the waitlist; no search or view tracking.

Physicians. Sorted by confirmed hours with a colour scale on that column: confirmed hours, visits, students, slots offered, fill rate, cancellations, waitlist, blocked, active since.

Students. Three tiles: activated students with zero visits; share of all visits taken by the most active fifth of students (a concentration measure); students blocked by a physician. A funnel: activated, booked, attended, logged hours, confirmed.

Recruit next. Collapsed to its top row with Show all. Every specialty with active physicians, open dates and waiting students, ranked by need: waitlisted students first, then no physician, then no open dates, then fewest open dates. Header: "N of M specialties have an active physician".

Term summary. One page for the current term, on demand and emailed as PDF to every administrator the day before the next term starts: confirmed hours, visits, students served, physicians active, top five physicians by hours, recruit-next line.

Exports

Export	Contents	Who
Table CSVs	The rows shown on Bookings, Physicians or Students	Data exports permission
Organization profile	Settings, terms, branding	Lead, step-up
Physician network	Physicians connected to this organization: contacts, specialties, locations, status	Lead, step-up
Shadowing history	Bookings, attendance, confirmed hours, cancellations	Lead, step-up
Student records	Participation and eligibility history	Lead, step-up
Analytics	Term summaries and aggregate reports	Lead, step-up
Program settings	Booking policies and configuration	Lead, step-up
Student's own	Hours log with reflections, CSV and PDF	The student

Reflections are never in an organization export. An organization export never includes another organization's students, bookings, preferences or relationships; for a shared physician it carries only what that organization is authorized to use. Package exports arrive as an emailed download link valid 24 hours.

10. Emails, notifications, and scheduled jobs

Forty-one message types. Every template uses the organization's name, logo, contact email and invitation link from the org record; nothing is hardcoded to one organization. Physician account mail (verification, welcome, confirm profile, regular summary, recommendation) is Keivos-branded and names an organization only where one is involved. Footers say "Reply to this email to reach your leadership" and replies route to the organization's contact email. Wording: "we'll notify you when they respond", never "either way"; never tell a student what leadership tracks; no patient information anywhere.

Routing. Confirmations and reminders go to the coordinators of the slot's location when it has any, else to the physician. The attendance question goes to whichever `attendance_to` names. Approval requests go to the physician and, where approval is delegated for that location, to its coordinators as well. Renewal, the regular summary, verification and account emails always go to the physician's professional email. Reminder timing (1, 2 or 7 days, or none), the summary cadence and renewal reminders are the physician's Preferences; booking requests, confirmations and attendance questions cannot be turned off.

Student

Email	Trigger	Actions
Your code	Sign-up verification or sign-in	None; code in the body, 15 minutes
Access request received	Request submitted	None
Access approved	Leadership approves, or a code is accepted after a request	Find a physician; welcome message included
Access not approved	Leadership denies	Reply, or request again
Access ends soon	Two weeks before `expires_at`	Enter the new code

Email	Trigger	Actions
Booking confirmed	Booking confirmed or approved	Add to calendar (.ics), Cancel (confirm page); site, address, go to, parking, day-of contact, before you come, cancel-by, paperwork
Request sent	Request joins the line	Place in line; Withdraw request
Request declined	Physician or delegated coordinator declines	Find another date
Reminder	cancel_hours before start	Cancel
How did it go	After the slot ends	Log hours and reflection
Nudge	3 days after, no log	Same
Hours confirmed	Physician answers Yes	None
Not confirmed	Physician answers No	Contact leadership (text)
Late cancellation recorded	Student cancels inside the window	Contact leadership (text)
Visit cancelled by physician	Physician or admin cancels	Pick a new date
A date opened	Waitlist policy fires	Book it
Extra visits granted or denied	Admin decides a cap request	Find a physician, if granted
Is your profile still right	Term start	Confirm, or update year and graduation
Request not answered in time	The visit starts with the request still in line	Find another date; says nothing is held against the student

Physician or scheduling coordinator

Email	Trigger	To	Actions
Organization invitation	Sent by the organization from its recruitment template (section 11); Keivos does not send it on its own	The physician or their office	Open the invitation link
Your code	Sign-in, identity verification or email change	The address being verified	None
Welcome to Keivos Shadowing	Account created (new profile or independent registration)	Physician	Open my portal; the emails to expect
Please confirm your shadowing profile	A coordinator prepares a profile for a professional email that has no account	The professional email the coordinator gave	Confirm (verifies the email, shows what was entered, confirms participation and the connection), Decline; one reminder at 7 days
Profile live	The organization approves the connection	Physician, and the location's coordinators	Open my portal
Approval request	Approve-each booking	Physician	Seat n of N and how many wait behind; student name, year, organization, bio and photo with contact details; View full profile; Approve, Decline (confirm pages). Sent when a request becomes the active one on a seat; never expires

Email	Trigger	To	Actions
Booking confirmed	Booking confirmed	Coordinators of the slot's location, else the physician	.ics; student contact details; location address
Reminder	reminder_days before, unless none	Same	Open my portal, Cancel (confirm page)
Did the student attend	Student logs hours	Per attendance_to	Yes, No (confirm pages); the No page offers a block
Attendance re-ask	7 days unanswered	Same	Same
Regular shadowing summary	On the physician's cadence (Monday for weekly and every 2 weeks, the 1st for monthly), only when the coming period holds visits, open requests or unanswered attendance, and the summary is on	Physician	Open my portal; a one-click action per item
Term renewal	3 weeks before an organization's term starts, when renewal_reminders is on; with reminders off the connection pauses at term start without mail	Physician	Same schedule (confirm page), Change or pause (portal)
Recommend Keivos	A physician recommends an organization	The named contact at that organization, physician copied	Learn more, Set up <organization>
Organization joined	A recommended organization signs up	The recommending physician	Connect with them

Email	Trigger	To	Actions
Visit went unfilled	A visit starts with requests still in line	Physician, and the location's coordinators	Informational: seats, how many were approved, how many requests were closed; distinguishes unfilled from partly filled; suggests turning off Approve each if unwanted
Coordinator invitation	A physician adds a coordinator to a location	The coordinator	Accept the invitation (sign in, or create the coordinator account with the email filled in)
Coordinator authorization request	A coordinator starts setup for a physician whose email already has an account	Physician	Approve (choose the location, whether they may approve requests), Decline; the coordinator is told only the outcome
You coordinate for <physician>	The physician approves a request, confirms a prepared profile, or changes what the coordinator may do	The coordinator	Open my coordinator portal

Administrators. Each administrator gets the digest on their chosen day (or never) and chooses which optional operational alerts to receive. Critical notices cannot be turned off.

Event	Default recipients	Optional
New physician connection (new profile, existing profile or confirmed coordinator setup)	Admins with Physicians	Yes

Event	Default recipients	Optional
Student access request	Admins with Students	Yes
No-show or late cancellation recorded	Admins with Students	Yes
Attendance unresolved 14 days	Admins with Physicians	Yes
Physician email delivery failure	Admins with Physicians	Yes
Cap increase requested	Admins with Students	Yes
Physician blocked a student	Admins with Students	Yes
Term summary	All administrators	Yes
Administrator invitation	The invitee and the leads	No
Permission or role change	The affected administrator and the leads	No
Sign-in from a new device	The administrator	No
Subscription or payment issue	Billing contact and the leads	No

Token rules

- Every action link lands on a page with one button; nothing is recorded by the click in the email, so mail scanners that pre-open links change nothing.
- The button submits a POST with the token. Single-use; a second visit shows "already answered" with the recorded answer and time.
- Expiry: attendance 30 days, approval until the visit starts, cancel until the slot starts, renewal until term start, profile link 90 days renewed by every email.
- Scope and role as in section 2.
- SPF, DKIM and DMARC on the sending subdomain before the first pilot email.

Scheduled jobs

- Nightly: slot generation; eligibility expiry by `expires_at`; graduation `access_until`; waitlist expiry at term end.
- Hourly: reminders, post-visit prompt and 3-day nudge, attendance re-ask at 7 days and admin alert at 14, automatic decline of requests still in line when a visit starts (by start time), with the student notices and the unfilled-visit summary, sequential waitlist offer expiry (commercial), billing retries (commercial).

- On event: waitlist notification when a slot opens; emails above.
- Calendar: digest per administrator day; physician summary on each physician's cadence (Monday weekly or every 2 weeks, the 1st monthly) when there is something to report; access-ends-soon two weeks before expiry; profile check at term start; term renewal 3 weeks before term start for physicians with reminders on, and auto-pause of unrenewed connections at term start; term summary the day before the next term starts; old activation code expiry at `code_expires`; coordinator-prepared profile reminder at 7 days and deletion at 30; expiry of unanswered coordinator authorization requests at 30 days.

11. Physician Recruitment Toolkit

Included with every commercial subscription and shown to the pilot under Physicians → Recruitment, beside the invitation link. For v1 it is a set of downloadable, editable resources, not an application module; the physician table already manages connected physicians, and a physician CRM is not built. Every resource carries the organization's invitation link.

Resource	Required behavior
Physician invitation email	Editable template with organization name, contact and invitation link filled in
Office manager outreach email	Alternative wording for scheduling staff
Alumni physician invitation	Emphasises the university connection
Phone outreach script	Short, customisable
Physician information sheet	Branded one-page PDF explaining participation
Recruitment tracker	Downloadable spreadsheet
Launch playbook	Recommended 30-day recruitment sequence
Follow-up templates	First follow-up, second follow-up, thank-you

Later: optional tracking of invitations and outreach status in Keivos.

12. Prerequisite Training Toolkit

A downloadable Canvas course package in .imscc format that an organization imports and customises: suggested lessons, assessments, acknowledgments and completion criteria. Versioned, so an organization knows which edition it runs; reviewed by clinical and educational personnel before distribution. Organizations can change the content, add local requirements and keep their own training policies, and are never required to use it. The toolkit is separate from eligibility verification: providing the course gives Keivos no access to the university's Canvas completion records.

13. Commercial onboarding, subscriptions, and refunds

Outside the pilot. How an organization becomes a customer:

1. An organization's representative creates an account.
2. Keivos verifies the organization's identity and university affiliation, assigns email domains and time zone.
3. The organization picks monthly or annual billing; auto-renewal is disclosed before purchase.
4. Payment is processed; the 90-day guarantee starts at the first successful payment, not at account creation.
5. The Lead Administrator works through a setup checklist: organization profile, booking rules, student access, academic terms, administrators.
6. The organization recruits physicians (section 11) and enrolls students.

Billing requirements. Subscription state (trialing, active, past_due, cancelled), renewal date, billing contact, invoices as PDF, payment-failure handling (retry, notify billing contact and leads, 14-day grace, then read-only), cancellation from Settings, refund processing.

Guarantee. One introductory 90-day guarantee per organization; full refund of payments made inside the window; no minimum recruitment or usage; a plain refund-request process; no deletion of organization data on refund; confirmed and pending visits are kept and either completed or cancelled with notice.

When a subscription ends. Confirmed visits before the end date are honoured; later ones are cancelled with the normal emails after a 14-day notice; the dashboard goes read-only; exports stay available for the retention period.

14. Data retention, privacy, and continuity commitments

An organization can leave Keivos without losing its own program records or the information it needs to continue its physician relationships, subject to privacy and authorization.

- The export package in section 9 is the mechanism; every item in it is available on request and on termination.
- Three kinds of data: the organization's program records (exportable), Keivos platform data (infrastructure, logs, the specialty list), and physicians' own account information (theirs; an organization receives only what it is authorized to use).
- Account closure preserves confirmed appointments until they are completed, cancelled or transferred; nothing is disabled with commitments unresolved.
- Commitments to publish, with legal review of the exact wording: transparent subscription pricing; no surprise charges for more students or physicians; advance notice of material pricing changes; paid periods are never repriced midterm; clear export rights; a change in Keivos ownership does not end customer agreements.
- Retention periods for active accounts, cancelled accounts, backups and legally required records are set with the infrastructure in hand (open decision 17), not invented here.
- Student reflections are private to the student in every case.

15. Infrastructure, security, and testing

Piece	Choice	Notes
App and API	shadow.keivos.com on the same hosting as the Advisor Dashboard and Student App	Student app, physician portal, coordinator portal, physician onboarding and admin are routes of one app; organization branding is a theme, not a site; Jared owns DNS
Database and auth	Existing Keivos Supabase project	RLS by org_id; physicians global with per-org links; emailed codes for every sign-in
Scheduled jobs	pg_cron or scheduled Edge Functions	Nightly, hourly and calendar jobs from section 10

Piece	Choice	Notes
Email	Existing provider on a dedicated subdomain	SPF, DKIM, DMARC; bounce webhooks write to email_log and alert admins
Token and code pages	Edge Functions or app routes	Stateless, look up the hash, record, render
File storage	Supabase storage	Logos, photos, paperwork PDFs, term summaries, exports
Billing	Payment provider (commercial release)	Webhooks drive subscription state
Monitoring	Sentry plus delivery webhooks	Failed jobs and bounces alert the responsible admins

Security

- Every permission check runs server-side; the client only hides what the server would refuse.
- Step-up code for lead-role grants, billing contact changes and package exports. Sign-in from a new device emails the administrator.
- Tokens hashed at rest, single-use, role-bound. Codes hashed, rate-limited, 5 attempts.
- Time zone is an explicit IANA zone per org set by Keivos, never the browser's.

Tests before the first real email

- Double booking: two bookings for the last seat in parallel; one must fail at the database.
- Overlap: an own rule overlapping a shared rule's slot is refused at save and by the exclusion constraint.
- DST: a slot either side of November 1, 2026 and March 14, 2027.
- Permission matrix: each permission group against each tab and action, including an admin with only Physicians never reaching student records or billing.
- Isolation: a physician connected to two organizations; organization A's export and screens contain nothing of organization B.
- Eligibility: expiry blocks new bookings, keeps existing ones flagged; term rollover does not revive an expired record.
- Email rendering in Gmail, Outlook and Apple Mail.

- Request line: three students request a one-seat visit in order; approving the first confirms seat 1 and leaves the others in line; cancelling the first makes the second the active request; the start-time job declines the rest and sends the summary. A declined student cannot request the same slot again.
- Overlap: a student with a confirmed 9 to 12 visit cannot request a 10 to 1 slot with another physician; an overlapping request that slipped through is refused at approval; two concurrent confirmations for overlapping slots cannot both succeed.
- Capacity: two concurrent approvals for the last seat; exactly one succeeds. Lowering capacity below the confirmed count is refused.
- Coordinator scope: a coordinator token issued for location A cannot act on a booking at location B; the coordinator portal renders location B's slots as Busy with no student data; a removed coordinator's next request is refused.

16. Release-specific acceptance checklists

Pilot: student

- [ ] Onboarding: profile with name parts and optional photo and bio, 6-digit email verification, activation code with inline error, Request access with note, awaiting-approval screen, welcome message on the done screen
- [ ] Sign in by code; pending and expired students land on the right screen
- [ ] Find a physician: Physicians and Availability views with the switcher remembered, filter box, cards, chronological date list, year-rule and block hiding, profile-confirm banner
- [ ] Physician detail: per-location address, logistics and paperwork; open dates with seats left and the booked, inside-notice, conflict, declined and in-line states; waitlist join and leave
- [ ] Confirm booking: rules list, acknowledgment; confirm, send request, or join the line with the place shown
- [ ] Monthly cap with inline Request more and the decision shown
- [ ] My shadowing: tiles, upcoming (confirmed) with Cancel and Cancel late (dialog with note), requests with place in line and Withdraw, history with log hours once and edit reflection, export CSV and PDF
- [ ] Account popover: profile, access line, sign out, rules and terms
- [ ] Blocked notice with the leadership contact; graduation and eligibility expiry close booking

Pilot: physician

- [ ] Invitation landing: Use my existing profile and Create a new profile, the coordinator line, the no-invitation line; the organization's description, student preparation and participation lists from Settings
- [ ] New profile: verify by code with an existing email recognised; professional profile with photo, name parts, MD/DO plus added credentials, Other specialty and bio; Connect; practice location with an optional coordinator invitation and paperwork; shadowing preferences; communication preferences; availability or finish later; Skip for now on every step after Connect; done screen with the setup checklist when anything was skipped
- [ ] Setup checklist on the portal Overview until location and shadowing preferences exist; gone afterwards even with no availability
- [ ] Existing profile: code, Connect with shared or own availability, shadowing preferences, pending connection for the administrator; the same from the portal's Connect box
- [ ] Coordinator-initiated setup: coordinator account by code, physician details, location, proposed availability; nothing visible until the physician answers; the physician's email never reveals whether an account exists; authorization request for an existing physician, confirmation for a new one; reminder at 7 days and deletion at 30
- [ ] Physician-initiated coordinator invitation from a location card and from onboarding; accept into an existing or new coordinator account
- [ ] Independent registration; Recommend Keivos to an Organization and its email
- [ ] Portal Overview: checklist, tiles (upcoming means confirmed), action cards with seat and line counts, approve, decline, attendance and renew
- [ ] Portal Schedule: calendar with locations and request counts, rules and dates with Applies to and Location, Edit and Remove, numeric students per visit, pause dialog, upcoming (confirmed) cards with cancel, request cards (active and in line), collapsed history with attendance and block
- [ ] Request line: approve fills a seat and promotes the next request; decline promotes; cancel of a confirmed student promotes; withdraw promotes; nothing expires on its own
- [ ] Portal My Organizations: per-organization settings, pause and resume, unblock, connect by invitation link or code
- [ ] Portal My Profile: name parts and credentials editor, email change by code, locations with add, edit, make default and the remove guard, coordinators per location (invite, let approve, stop approving, remove), email preferences, activity

- [ ] Sign-in by code; forwarded links act as the recipient and open no session; a coordinator token outside its location is refused
- [ ] Confirmation dialogs on cancel, remove, block, decline, no-show, location remove and coordinator remove
- [ ] Token confirm pages: Yes, No (with optional block), Cancel, Approve, Decline, Same schedule, Confirm profile, Authorize coordinator, already answered, expired

Pilot: scheduling coordinator

- [ ] Account creation by code; sign-in by code; Accept on a pending invitation
- [ ] Overview across physicians: tiles, delegated requests, attendance questions, upcoming cards, the note about physicians who approve personally
- [ ] Schedule with the physician selector, full calendar, other locations as Busy with no student data, rule lists with other locations marked and without controls
- [ ] Availability: rules and dates limited to authorized locations, attributed to the coordinator in the physician's activity
- [ ] Visits & Requests: approve and decline only when delegated, attendance only when the physician routes it to the coordinator, cancel always, collapsed history for their locations
- [ ] Practice locations: edit logistics and paperwork; no default, remove or coordinator changes
- [ ] Account: name parts, phone, authorizations list; no access to the physician's profile, organizations, coordinators or email

Pilot: administration

- [ ] Sign in by code; invitation accepted by signing in
- [ ] Lead and Organization Administrators; permission groups enforced server-side; tabs and settings hidden accordingly; last-lead guard; step-up for lead grants, billing contact and exports
- [ ] Overview tiles and action items incl. access requests
- [ ] Physicians tab: Physicians and Recruitment views; table with Site, Available, Booked and Joined; approve, pause dialog, resume; read-only profile with tabs Profile, Locations, Availability, Students, Blocked, Activity; Recruitment view with the invitation link and outreach resources
- [ ] Bookings with period filter, cancel with confirmation, CSV
- [ ] Students with eligibility column and filters; actions panel incl. approve, deny, revoke, reinstate, graduate

- [ ] Analytics: physicians, students, Recruit next; term summary on demand and as PDF
- [ ] Settings: all six sections; locked Keivos settings; branding applied to the student app and emails
- [ ] Every table sortable and filterable

Pilot: jobs and email

- [ ] Nightly slot generation and eligibility expiry; hourly reminders, prompts, re-asks, expiries; automatic decline of requests in line at visit start with the student notices and the unfilled summary
- [ ] Digest per administrator day; optional alerts honoured; critical notices always sent
- [ ] Access-ends-soon, profile check, term renewal per physician preference and auto-pause, regular physician summary on its cadence, staff-setup reminder and deletion, term summary
- [ ] All templates use org variables; rendered in Gmail, Outlook and Apple Mail
- [ ] SPF, DKIM, DMARC (Jared adds the records); the tests in section 15 pass

Commercial release (in addition)

- [ ] Self-service organization sign-up with Keivos verification; setup checklist
- [ ] Billing: plans, invoices, payment failure handling, cancellation, refunds, 90-day guarantee, auto-renew disclosure
- [ ] Sequential waitlist offers with response window, maximum offers and notice-deadline release
- [ ] Canvas automatic verification, if proven
- [ ] Recruitment toolkit and training toolkit downloads
- [ ] Export package with step-up; retention periods implemented; account closure flow
- [ ] Published business terms (section 14)

17. Deferred features and unresolved decisions

Deferred

- Canvas OAuth or LTI 1.3 verification once feasibility and API access are tested.
- Sequential waitlist offers (commercial release).
- Outreach tracking inside Keivos; recommendation tracking beyond the sent and joined flags; Keivos admin tools to merge specialties added through Other.
- Response-time metric per physician.

- Letter-of-recommendation tracking, physician services marketplace, LMS, LTI launch, roster imports: not planned.

Open decisions

- [ ] Automatic Canvas verification: validate OAuth or LTI feasibility before choosing an implementation.
- [ ] Sequential waitlist offers: first commercial release, or later.
- [ ] Graduation access: the maximum extension in months, and how graduates sign in once they lose school email (leadership-assisted account recovery is the working assumption).
- [ ] Faculty adviser designation: informational only, or also required for account recovery.
- [ ] Retention periods for active, cancelled, backup and legally required records, and the export window after cancellation.
- [ ] Price protection and the other customer-facing commitments: final wording, reviewed before publication.
- [ ] Daily currently means Monday to Friday; confirm or add a seven-day option.
- [ ] Keep the "most active fifth of students" tile, or drop it.
- [ ] Capacity ceiling: 50 students per visit in the mock-ups as a technical limit; confirm the backend maximum. Student bios: no moderation in v1, leadership can ask a student to change one; decide whether administrators see bios. Coordinators: whether a physician may grant every location at once, and whether a coordinator should see the names of students at locations they are not authorized for (v1: busy time only).
- [ ] Which locations an organization sees: every location that has dates (the v1 assumption), or let a physician limit a location to certain organizations.
- [ ] Blocks are physician-wide and the portal lists each blocked student under the organization they belong to; confirm that no organization is told about a block made through another.
- [ ] Time zones when a physician's location sits in a different zone from the organization: the location's zone for the physician, org.timezone for students (the v1 assumption).
- [ ] Staff-prepared accounts: one reminder at 7 days and deletion at 30 (the v1 assumption).
- [ ] Independent registrations with no connection after a year: keep, remind, or archive.

18. Mock-up placeholders

Everything below is hardcoded or faked in the six HTML mock-ups and must come from the database, the signed-in session or a job in the real build.

Mock-up value	Real source
The clock is fixed at October 25, 2026 (`TODAY`, `TODAY_ISO`); every relative date, notice window, cancel-by time and period is computed from it	Server time in org.timezone or the location's zone
Org config: organization name and university, slug and invitation link, logo (none), accent colour, welcome message, program description, student preparation and participation lists, contact email, time zone, byui.edu, 72-hour cancellation, 2 visits a month, waitlist, graduation and renewal policies, access requests on, code F26-SHDW-4K2 expiring Jan 4 2027, pilot plan, five terms	orgs, terms, subscriptions
Four administrators with roles, permissions, adviser flag, invited status, digest days and alert choices; the signed-in admin is Brother Shaw, a lead	org_admins and the session
Signed-in identities: student Avery Jensen, physician Dr. Elena Marsh, admin Brother Shaw	The session
Six sample physicians with invented .example emails, practice locations (Dr. Marsh has two, her profile set up by her coordinator), rules and dates pointing at locations (one date offered to the Idaho State organization alone, hidden from BYU-Idaho screens), summary cadences, activity logs, paperwork file names; Dr. Marsh connected to an invented Idaho State organization	physicians, practice_locations, physician_orgs, availability_rules, manual_dates, activity_log, storage

Mock-up value	Real source
Six sample students with eligibility records (one pending access request, one expired), seventeen bookings across three terms, one waitlist entry, one cap request, one blocked student	users, eligibility_records, bookings, visit_logs, waitlist, cap_requests
The 45-entry specialty list; Other adds to it in memory only	specialties, Keivos-wide
Any six digits pass email verification, sign-in and step-up; the activation code must match the config string; Use my existing profile and the portal's Connect box always resolve to Dr. Marsh; a new or independent profile lives in page memory; permission checks only hide controls	email_codes with hashed codes, attempts and expiry; real account lookup; server-side permission enforcement
Coordinator-initiated setup ends at a done screen and the physician's confirmation and authorization emails exist only on the Emails page; Recommend Keivos, Connect with an organization and Open my portal show a toast	Staff-prepared accounts with confirmation tokens; org_recommendations and the recommendation email; a real portal session
Every email uses sample recipients, dates and names; confirm pages change state in the page only	Templates rendered per event; tokens recorded server-side
"Renew for Winter 2027" and the renewal email name the next term as a literal; the summary email shows one sample period	terms table; the Monday job
Pause-until default Jan 11 2027; new individual date default Dec 12 2026; calendar opens on November 2026; new rule defaults (Weekly, Tuesday, 8 to 12, 14-day notice, 60 days ahead, 1 student, the default location)	Defaults relative to today; rule defaults can stay as product defaults

Mock-up value	Real source
Copy buttons use the clipboard; Export, Download PDF, Resend, outreach Edit and Download, export package and billing contact change only show a toast after the step-up	Real files, email resend, billing provider
Photo and logo uploads read into memory as data URLs; paperwork uploads keep file names only; the accent colour is stored but not applied to the theme; a location's time zone is not shown	Supabase storage; a theme applied from org.accent_color with the contrast rule; timezone on practice_locations
Analytics counts, open-date counts, fill rates, the funnel and Recruit next are computed in the browser from sample data	SQL views over bookings, slots, visit_logs, waitlist
The term summary's "emailed on" date is computed from the next term; nothing is sent	Scheduled job
Sequential waitlist offers, Canvas verification and invoices appear only as greyed options or empty rows	Commercial release
Every page loses state on reload; there is no server, auth, persistence, rate limit or RLS; time zone is the browser's	All of it
Request lines: Noah is the active request and Maya is #2 on Dr. Marsh's Dec 8 date; Dr. Marsh routes attendance questions to her coordinator; Dr. Raman's Nov 12 date (2 seats) conflicts with Avery's confirmed Nov 12 visit; student bios for Noah and Maya; two sample coordinators (Teresa Oakes for Dr. Marsh and Dr. Ahn, Dana Ferris for Dr. Raman); names stored as parts and phones as E.164 in the sample data	bookings with queue_position and seat, users.bio, coordinators, coordinator_authorizations

Mock-up value	Real source
Approve, decline, cancel and withdraw move the next request up inside the page; the automatic decline at visit start, the unfilled summary and the overlap check at approval never run; Skip for now, the setup checklist and coordinator invitations live in page memory	The slot transaction, the start-time job, the exclusion constraints
The coordinator portal is signed in as Teresa Oakes; Create a coordinator account, Accept on a pending invitation and the coordinator-initiated setup change state in the page only and never check whether an email has an account	coordinators, coordinator_authorizations, email_codes, the private account check